



AI Hackathon Challenge Statement

Real-Time Vietnamese-English Business Meeting Translator Challenge |
Sponsored by AI Singapore

Challenge Overview

In this 2-day AI hackathon, teams are challenged to build a working prototype of a real-time translator for in-person business meetings between Vietnamese and Singaporean delegations. The solution must support Vietnamese-to-English and English-to-Vietnamese communication with near real-time performance, enabling both sides to communicate accurately, efficiently, and with a smooth user experience. The expected outcome is a working demo that can be used live during the judging session, where judges and participants will engage in free-flow exchanges in both Vietnamese and English.

The prototype may run on a laptop, tablet, smartphone, or another small device assembled during the competition. The challenge does not prescribe whether translated output must be delivered as voice or text; either is acceptable, provided the result appears almost in real time and is practical for live meeting use.

Core Requirements

- Build a functional prototype within the 2-day hackathon period.
- Support bidirectional translation between Vietnamese and English.
- Operate in a live, in-person business meeting setting.
- Be demonstrated live during the judging session through free-flow exchanges in Vietnamese and English between the judges and the participants.
- Deliver translations with strong communication accuracy and low perceived latency.
- Provide a user experience that is intuitive, efficient, and minimally disruptive to the meeting flow.
- Run on accessible computing hardware such as a laptop, tablet, smartphone, or a compact custom-built device.
- Teams may use AI assistants to help design, code, test, and refine their solution.



Team Rules

Each competing team may consist of up to three engineers and/or scientists.

Bonus Considerations

- Bonus points will be awarded for solutions built using open AI models that can be hosted on-premise, making them suitable for meetings involving confidential or sensitive information.
- Additional bonus points will be awarded for solutions that can be deployed on edge devices.
- Bonus points will also be awarded for working demos that remain effective in noisy environments and handle conversational turn-taking well.
- Teams will also be recognized for designing extensible systems that can be adapted to additional language pairs, especially lower-resource languages.

What Strong Solutions Should Demonstrate

Strong submissions should demonstrate a clear balance between translation quality, responsiveness, usability, and deployability in realistic business settings. Teams are encouraged to think beyond raw model performance and consider practical issues such as turn-taking, background noise, speaker changes, privacy, hardware constraints, and how their interface supports natural conversation flow.

Judging Rubric

Judges may assess each team based on the following criteria, using the suggested weightings below to guide overall scoring.

Criterion	Suggested Weight	What Judges Should Look For
Translation accuracy	30%	How accurately the solution preserves meaning, intent, and key details in both Vietnamese-to-English and English-to-Vietnamese exchanges.
Latency and responsiveness	20%	How quickly the system produces output and whether it supports a natural, near real-time conversational flow.
User experience and meeting flow	20%	How intuitive, smooth, and minimally disruptive the interaction is during a live business-style conversation.



Robustness in realistic conditions	15%	How well the system handles turn-taking, speaker changes, and environmental noise during the live demo.
Technical design and deployability	15%	The soundness of the implementation, suitability for practical deployment, and clarity of the overall system design.

In addition to the base rubric, judges may award bonus points for on-premise use of open AI models, edge-device deployment, strong performance in noisy environments, effective turn-taking support, and extensibility to other language pairs, especially lower-resource languages.

Prize and Opportunity

This challenge is sponsored by **AI Singapore**. The winning team will receive prize money, as well as the opportunity for a sponsored 1- to 2-month visiting researcher experience at the AI Singapore office at Nanyang Technological University in Singapore.

This challenge invites participants to combine speech, translation, systems engineering, and human-centered design to create a tool that can make multilingual business communication more seamless, private, and inclusive.